Pass-Through, Welfare, and Incidence under Product Returns in Perfect Competition

Draft notes

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1 Introduction

About one in five products sold online is returned. Pass-through, tax incidence, and the efficiency cost of taxation are well understood when purchase settles who keeps the good. In that classical setting, the demand and supply curves are sufficient for consumer surplus, producer surplus, and deadweight loss. With returns, allocation is decided in two stages. Buyers first choose whether to purchase, then whether to keep or return after receiving the item. Never buying and buying then returning leave the same final allocation—the consumer does not keep the good—yet their welfare costs differ. A returned purchase incurs shipping, hassle, and handling that a non-purchase avoids, so demand and supply alone omit an economically relevant margin.

What objects then replace the classical demand and supply curves, and how misleading is an analysis that ignores returns? We study these questions under perfect competition with product returns and seller handling costs. We characterize pass-through, incidence, and welfare losses with returns, compare them to the no-return benchmark, and identify where omitting the return margin changes conclusions.

Even with a return margin, surplus continues to rest on a small set of market aggregates. What changes is which aggregates matter and at which prices they are evaluated. Consumer surplus depends only on two aggregates, gross demand and the return rate. Heterogeneity in tastes, match values, and return costs enters welfare solely through these aggregates. Producer surplus retains the usual geometry—area to the left of the supply curve—but is evaluated at the effective margin, list price minus expected refunds and handling, rather than at list price.

Because consumers pay the list price while firms respond to the effective margin, pass-through from cost shocks into list prices is governed by how that margin moves with the list price. The response of the effective margin to the list price reflects two forces. First, a higher price directly raises revenue on kept units. Second, the return rate itself may change with price, which alters the expected refund and handling costs the seller incurs. Whether the return rate rises or falls depends on a behavioral channel—higher prices make consumers more likely to return—and a selection channel—price changes alter the composition of buyers. When the behavioral channel dominates, a higher list price actually lowers the effective margin, making supply slope downward in the list price and setting the stage for the surprising pass-through results below.

A related and equally important departure concerns tax incidence. In the standard model, it does not matter which party remits a tax. With returns, this neutrality fails because the refund price differs by remittance regime. Under seller remittance, the tax-inclusive checkout price is refunded. Under buyer remittance, only the pre-tax list price is. Since the return decision depends on the refund price, the two regimes generate different effective margins and equilibria. These two departures—the behavioral response of returns to price and the failure of tax neutrality—are the central forces that shape the pass-through results below.

We show that pass-through of a one-dollar production-cost shock equals that of a seller-remitted unit tax, but differs under buyer remittance. The ratio of handling to production-cost pass-through equals the aggregate return rate. When the behavioral response of returns to price is strong enough that the effective margin falls in list price, buyer- and seller-remitted pass-through carry opposite signs, a sharp reversal of the classical neutrality intuition. Which regime yields a negative pass-through depends on the relative magnitudes of the supply and demand elasticities. Intuitively, negative pass-through arises when the list-price supply curve slopes downward and more steeply than demand. A higher price reduces the effective margin and hence quantity supplied. In this region, a cost increase creates excess demand, and the equilibrium price must fall to restore market clearing.

A striking result emerges from the incidence analysis. Despite pass-through of a cost shock potentially being negative, the incidence ratio is always positive and identical across all four shocks—production cost, handling cost, and both seller- and buyer-remitted taxes. This uniformity implies that consumer surplus, producer surplus, and welfare always move in the same direction, similar to the classical model without returns. Nevertheless, the welfare implications can be very different and hinge critically on the sign of pass-through. When cost pass-through is negative, production cost, handling cost, and seller tax shocks increase both surpluses, yielding a Pareto improvement. For buyer-remitted taxes, both surpluses unambiguously decline. When pass-through is positive, all four shocks reduce both surpluses. The two tax regimes therefore generate opposite welfare outcomes precisely when pass-through is negative, just as their pass-through signs can oppose each other.

These welfare effects also reshape first-order private-surplus losses. Without returns, a unit production-cost or tax increase lowers private surplus by exactly the market quantity. With returns, that classical loss is scaled by a factor that reflects the price-induced reallocation between kept and returned units and the associated handling waste. When pass-through is positive and returns rise with price, the factor exceeds one, so the no-return formula underestimates the welfare loss. When returns fall with price, the factor drops below one and the classical formula overestimates the loss. When pass-through is negative, the factor itself changes sign. A cost or seller-tax increase then raises private surplus—the same Pareto improvement as above—because the induced fall in list price reduces costly returns by enough to more than offset the direct resource drain. An analyst who ignores returns then not only mismeasures the magnitude but gets the wrong sign.

To sign that bias, we provide a simple diagnostic, from the return-rate elasticity and the ratio of handling to list price, for when the classical no-return pass-through formula over- or underestimates the correct returns-adjusted pass-through. Parallel conditions in the return-rate, supply, and demand elasticities sign when the classical welfare formula over- or understates the true first-order welfare change, or reverses its sign. This stands in sharp contrast to the classical no-return benchmark, where pass-through is always positive and such welfare-improving outcomes are impossible.

2 Setting

We begin with the market primitives: purchase and return behavior on the consumer side, and competitive supply with return handling on the firm side. A homogeneous product is sold at list price p. Consumers indexed by i differ in return hassle κ_i, the disutility from returning the product, and in match distribution G_i over post-purchase match quality u. After purchase, type i draws $u \sim G_i$ and keeps the item whenever $u - p \geq -\kappa_i$, earning surplus $u - p$, and returns it otherwise, earning $-\kappa_i$.

Buying at price p yields expected surplus

$$cs_i(p) = \mathbb{E}_{u\sim G_i}\big[\max(u-p,\, -\kappa_i)\big].$$

On the purchase margin, all consumer-side heterogeneity is summarized by willingness to pay v_i, defined by $cs_i(v_i) = 0$, so type i buys if and only if $p \le v_i$. Across types, willingness to pay has distribution $F(v) = \Pr(v_i \le v)$ with density f, so gross demand is $D(p) = 1 - F(p)$. Of the units sold at price p, a share $r(p)$ are returned.

To understand how $r(p)$ changes with price, write type i's return probability as $\rho_i(p) = \Pr(u < p - \kappa_i)$. Willingness to pay is sufficient for purchase but not for returns: types with the same v_i may still differ in (κ_i, G_i). Averaging within WTP cells,

$$\rho(v,p) \equiv \mathbb{E}\big[\rho_i(p) \bigm| v_i = v\big],$$

so the aggregate return rate is the truncated mean

$$r(p) = \mathbb{E}\big[\rho(v_i,p) \bigm| v_i \ge p\big].$$

Differentiating yields

$$r'(p) = \mathbb{E}\big[\partial\rho(v_i,p)/\partial p \bigm| v_i \ge p\big] + \frac{f(p)}{D(p)}\big(r(p) - \rho(p,p)\big),$$

where $\rho(p,p)$ is the average return probability among types at the purchase margin $v_i = p$.[1] The first term is behavioral: a higher keep threshold raises return probabilities among those who still buy. The second is selection: a higher price drops cutoff types at the purchase margin, and that term would be negative if those cutoff types return more than the average purchaser.[2] Behavioral and selection effects can therefore pull in opposite directions, and we do not impose a sign on $r'(p)$.

The product market is perfectly competitive. Producer j takes p as given, chooses output at cost $C_j(q_j)$, and incurs a net return handling cost h on each returned unit, equal to gross handling cost minus the scrap value of the returned product. This net cost may be negative, but we maintain $p + h > 0$, which holds whenever scrap value does not exceed list price.

3 Consumer side

We turn first to consumer surplus and show that only the aggregates D and r enter. Write $cs(v,p)$ for expected surplus at list price p among types with willingness to pay v, so $cs(v,v) = 0$. Consumer surplus integrates surplus over all types who buy,

$$CS(p) = \int_p^\infty cs(v,p)\, dF(v). \tag{1}$$

[1] With $\rho(v,p)$ as defined in the text, $r(p) = D(p)^{-1}\int_p^\infty \rho(v,p)\,dF(v)$. Differentiating and using $D'(p) = -f(p)$ gives

$$r'(p) = D(p)^{-1}\int_p^\infty \partial\rho(v,p)/\partial p\, dF(v) + \frac{f(p)}{D(p)}\big(r(p) - \rho(p,p)\big),$$

which is the display in the text.

[2] Selection need not be negative for arbitrary (κ_i, G_i). It is signed under an additive match specification $u = v_i + \varepsilon$ with shock ε of any distribution independent of willingness to pay, and with return hassle likewise independent of v: then $\rho(v,p)$ decreases in v, so $\rho(p,p) \ge r(p)$ and the selection term is non-positive.

Differentiating in p gives an extensive-margin term from types entering or leaving the purchase margin and an intensive-margin term from lower surplus among inframarginal buyers,

$$CS'(p) = -cs(p,p)\, f(p) + \int_p^{\infty} cs_p(v,p)\, dF(v),$$

where $cs_p(v,p) \equiv \partial cs(v,p)/\partial p$. On the extensive margin, $cs(p,p) = 0$, so entry and exit contribute nothing at first order. On the intensive margin, $cs_p(v,p)$ equals minus the keep probability, because a higher price harms the consumer only when she keeps the purchase,

$$cs_p(v,p) = -\Pr(u \geq p - \kappa_i).$$

Aggregating over buyers who purchase then gives

$$CS'(p) = \int_p^{\infty} cs_p(v,p)\, dF(v) = -D(p)\big(1 - r(p)\big),$$

by the definitions of $D(p)$ and $r(p)$. A small price change therefore satisfies

$$dCS = -D(p)\big(1 - r(p)\big)\, dp. \tag{2}$$

The extensive- and intensive-margin decomposition makes the local formula transparent. Consumer surplus changes depend only on the aggregates $D(p)$ and $r(p)$, both at each price and integrated along any path,

$$\Delta CS = -\int_{p_0}^{p_1} D(p)\big(1 - r(p)\big)\, dp.$$

Micro heterogeneity in price responses can be rich. Price p enters each $cs_i(p)$ nonlinearly and can shift participation, return decisions, and inframarginal surplus differently across types. Yet once (D, r) are given, this microstructure is irrelevant for dCS and ΔCS. Two economies with the same paths of $D(p)$ and $r(p)$ deliver the same consumer surplus loss, whether high-v_i types adjust return behavior more steeply than low-v_i types or not. Only kept units enter at first order, through $1 - r(p)$, not gross purchases $D(p)$ alone.

Let $\bar{p}$ denote the highest willingness to pay in the population, so $D(\bar{p}) = 0$ and $CS(\bar{p}) = 0$. Integrating the local formula then rewrites consumer surplus as area under kept volume,

$$CS(p) = \int_p^{\bar{p}} D(z)\big(1 - r(z)\big)\, dz. \tag{3}$$

4 Supply side

On the firm side, returns make list price an imperfect summary of producer payoffs. A gross sale brings revenue p if the consumer keeps the item and costs refund p plus net handling h if the item is returned. Among gross sales at list price p, share $r(p)$ are returned, so expected revenue per gross unit—the *effective margin*—is

$$\mu(p,h) \equiv p - r(p)(p + h). \tag{4}$$

Under perfect competition, firm j takes p as given and therefore faces a fixed number μ in its supply decision. At equilibrium, $\mu = \mu(p,h)$.

Given μ, firm j chooses output to solve

$$\max_{q_j \geq 0} \;\; \mu\, q_j - C_j(q_j), \qquad \text{FOC: } C_j'(q_j) = \mu \;\; (\text{if } q_j > 0),$$

with $C_j(0) = 0$. Write $q_j(\mu)$ for firm j's supply, set to zero when μ is below its minimum marginal cost. We assume each C_j has strictly increasing marginal cost, so $q_j'(\mu) > 0$ for active firms.

Summing across firms defines market supply and industry profit,

$$S(\mu) \equiv \sum_j q_j(\mu), \qquad PS(\mu) \equiv \sum_j \big[\mu\, q_j(\mu) - C_j\big(q_j(\mu)\big)\big]. \tag{5}$$

Under that assumption, $S'(\mu) > 0$. Market supply therefore rises with the effective margin, but not necessarily with list price. Firms care about μ, and $\mu(p,h)$ can fail to rise with p when refunds and handling eat into the list-price gain. Which firms are active at each μ determines both the market quantity and total profit.

Differentiating $PS(\mu)$ with respect to μ gives, for each firm,

$$\frac{d}{d\mu}\Big[\mu\, q_j - C_j(q_j)\Big] = q_j + q_j'(\mu)\,\big[\mu - C_j'(q_j)\big].$$

Active firms satisfy $C_j'(q_j) = \mu$ at their chosen output, so the adjustment term $q_j'(\mu)\,[\mu - C_j'(q_j)]$ is zero for each firm—equivalently, the envelope theorem applied to each firm's profit problem gives $d\pi_j/d\mu = q_j(\mu)$. Summing over firms,

$$PS'(\mu) = \sum_j q_j(\mu) = S(\mu).$$

With no production at $\mu = 0$, it follows that industry profit equals the area to the left of the market supply curve,

$$PS(\mu) = \int_0^{\mu} S(z)\, dz. \tag{6}$$

At equilibrium $\mu = \mu(p,h)$, $PS = \int_0^{\mu(p,h)} S(z)\, dz$.

5 Failure of tax neutrality

With firms responding to μ rather than list price, a natural question is whether remittance of a unit tax remains neutral. Weyl and Fabinger (2013) show that without returns, buyer and seller remittance deliver the same equilibrium. With returns, we ask whether that equivalence survives by writing the two clearing conditions at a common checkout price p—the total payment at the register—and a per-unit tax τ. Fix p and τ. Seller remittance clears at

$$D(p) = S\big(p - \tau - r(p)(p+h)\big), \tag{7}$$

because both the return rate and per-return cost $p + h$ are evaluated at checkout price p. Buyer remittance clears at

$$D(p) = S\big((p-\tau) - r(p-\tau)(p-\tau+h)\big), \tag{8}$$

because the seller posts list price $p - \tau$, so returns and refund cost are evaluated there. The two conditions define the same equilibrium only if the arguments of $S(\cdot)$ coincide. Canceling $p-\tau$ yields

$$r(p)(p+h) = r(p-\tau)(p-\tau+h). \tag{9}$$

Result 1 (Failure of tax neutrality with returns)**.** *Seller and buyer taxes define the same equilibrium if and only if* $r(p)(p+h) = r(p-\tau)(p-\tau+h)$.[3]

Tax neutrality breaks down because the refunded amount—and therefore the keep-or-return decision—depends on which party remits the tax. Under seller remittance the tax-inclusive checkout price is refunded; under buyer remittance only the pre-tax list price is. Purchase is always based on checkout price p, but returns are evaluated at p under seller remittance and at $p-\tau$ under buyer remittance, so the effective margin generally differs and the two clearing conditions are not equivalent.

6 Pass-through

Equilibrium satisfies $D(p) = S(\mu(p,h))$. Write $Q \equiv D(p)$. We characterize pass-through into list price for three shocks: a uniform one-dollar parallel shift in every firm's marginal production cost (indexed by c), a one-dollar increase in handling cost h, and a unit tax under seller and buyer remittance. Production cost shifts market supply at a given effective margin μ, while handling lowers $\mu(p,h)$ at a fixed list price p.

For production and handling, write the clearing condition as

$$m(p,x) \equiv D(p) - S(\mu(p,x),x) = 0, \qquad x \in \{c,h\}. \tag{10}$$

Differentiating while keeping $m = 0$ defines pass-through $\rho_x \equiv dp/dx$ by

$$\rho_x = -\frac{m_x}{m_p}, \tag{11}$$

and expanding $m = D - S$ gives

$$m_p = D'(p) - S'(\mu)\,\mu_p, \qquad\qquad m_x = -S'(\mu)\,\mu_x - S_x, \tag{12}$$

hence

$$\rho_x = \frac{-S'(\mu)\,\mu_x - S_x}{S'(\mu)\,\mu_p - D'(p)}. \tag{13}$$

The two shocks differ in which margin moves at fixed p or fixed μ,

$$\mu_c = 0, \qquad S_c = -S'(\mu), \qquad \mu_h = -r(p), \qquad S_h = 0. \tag{14}$$

The production term is $S_c = -S'(\mu)$ because a one-dollar parallel rise in every firm's marginal cost is equivalent, at fixed μ, to evaluating market supply at $\mu - 1$. Handling moves μ by only $r(p)$ per dollar relative to that unit production shift, so its numerator in (13) is the fraction $r(p)$ of the production numerator.

Turning to the unit tax, failure of tax neutrality requires separate checkout prices p^s and p^b under seller and buyer remittance. Write $\rho^s_\tau \equiv dp^s/d\tau$ and $\rho^b_\tau \equiv dp^b/d\tau$. At $\tau = 0$, (13) with $S_\tau = 0$ gives

$$\rho^s_\tau = \frac{-S'(\mu)\,\mu^s_\tau}{S'(\mu)\,\mu_p - D'(p^s)}, \qquad \rho^b_\tau = \frac{-S'(\mu)\,\mu^b_\tau}{S'(\mu)\,\mu_p - D'(p^b)}. \tag{15}$$

[3] Neutrality requires the return wedge to be the same at checkout price p and at posted list price $p-\tau$. That holds for all taxes only if $r(p)(p+h)$ is constant in posted list price, which requires $r(p) = C/(p+h)$ for some constant C; without that functional form on r, the two clearing conditions need not define the same equilibrium.

The regimes differ only through μ^s_τ and μ^b_τ at a fixed checkout price. At $\tau = 0$, $\mu^s_\tau = -1$. Under buyer remittance, τ enters the effective margin in the same way as $-p$: a tax increase lowers the pre-tax price entering $\mu(p-\tau, h)$ by one dollar, changing μ by $-\mu_p$. Hence $\mu^b_\tau = -\mu_p$ and $\rho^b_\tau = \mu_p\,\rho_c$. The sign of buyer-remitted pass-through is thus tied to that of seller-remitted pass-through by the sign of μ_p. When returns are absent ($r = 0$), $\mu_p = 1$ and the two pass-through rates coincide, recovering the classical neutrality benchmark. If $\mu_p < 0$, however, the two rates always carry opposite signs—a striking reversal of that neutrality intuition.

Result 2 (Pass-through into list price and remittance-dependent sign)**.** ***(i)*** *Production-cost pass-through is*

$$\rho_c = \frac{S'(\mu)}{S'(\mu)\,\mu_p - D'(p)}. \tag{16}$$

The remaining shocks connect to this object by

$$\rho_h = r(p)\,\rho_c, \qquad \rho^s_\tau = \rho_c, \qquad \rho^b_\tau = \mu_p\,\rho_c. \tag{17}$$

(ii) *If* $\mu_p < 0$*, then*

$$\operatorname{sgn}(\rho^b_\tau) = -\operatorname{sgn}(\rho_c)$$

provided $\rho_c \neq 0$*. A unit tax therefore moves the equilibrium checkout price in opposite directions depending on which side of the market remits it.*

The condition $\mu_p < 0$ requires the behavioural response of returns to price to be strong enough that the effective margin falls when p rises: from $\mu_p = 1 - r(p) - r'(p)(p + h)$, we have $\mu_p < 0$ exactly when $r'(p)(p+h) > 1 - r(p)$. In that case buyer remittance raises μ while seller remittance lowers it, shifting supply outward and inward respectively. The checkout price must therefore adjust in opposite directions to restore equilibrium. Which regime raises the price and which lowers it depends on the sign of ρ_c, which in turn hinges on the relative magnitudes of the supply and demand elasticities—a point developed in the next subsection.

6.1 Elasticity representation

The same pass-through formulas can be rewritten in elasticities, which makes the role of μ_p transparent. Write $\mu_p \equiv d\mu/dp$ for the response of the effective margin to list price, and define demand and supply elasticities with respect to list price as usual by

$$\varepsilon_D(p) \equiv \frac{pD'(p)}{D(p)} < 0, \qquad \varepsilon_S(p) \equiv \frac{dS(\mu(p))}{dp}\,\frac{p}{S(\mu(p))} = S'(\mu)\,\mu_p\,\frac{p}{S(\mu)}.$$

Here $\varepsilon_S(p)$ is the observed elasticity of market supply with respect to the clearing price. Without returns, $\mu_p = 1$ and $\varepsilon_S > 0$. With returns, μ_p can become negative, in which case $\varepsilon_S < 0$ and a higher list price reduces the quantity firms are willing to supply.

Without returns, supply is a function of list price, so $S' = dS/dp$. A one-dollar cost increase cuts supply by S' units at the original price, creating an excess-demand gap of that size. Raising list price by one dollar closes the gap from both sides. Demand falls by $|D'|$ and supply recovers by S'. Closing a gap of size S' therefore requires a price rise of $S'/(S' + |D'|)$, so standard pass-through equals $\varepsilon_S/(-\varepsilon_D + \varepsilon_S)$.

With returns, supply is a function of the effective margin, so $S'(\mu) = dS/d\mu$ rather than dS/dp. A one-dollar production-cost increase is equivalent to lowering μ by one dollar, so at fixed list price supply falls by $S'(\mu)$ units, again creating an excess-demand gap of that size. Raising list price by one dollar closes the gap at rate $S'(\mu)\,\mu_p - D'(p)$, so

$$\rho_c = \frac{S'(\mu)}{S'(\mu)\,\mu_p - D'(p)}.$$

The cost shock lowers μ by one dollar, while a dollar of list price raises μ by μ_p. A higher μ_p means a one-dollar price increase restores more of the effective margin—and therefore more quantity supplied—than the one-dollar cost shock removes, so a smaller list-price movement closes the excess-demand gap. Measuring supply against list price through ε_S puts the factor $1/\mu_p$ in the numerator,

$$\rho_c = \frac{\varepsilon_S(p)/\mu_p}{-\varepsilon_D(p) + \varepsilon_S(p)}. \tag{18}$$

When $\mu_p < 0$, the supply curve expressed as a function of the list price p is locally decreasing. In that region the sign of ρ_c is no longer guaranteed positive. It depends on the relative magnitudes of ε_S and ε_D. This opens the door to the counterintuitive outcome that an increase in marginal cost reduces the equilibrium price.

Result 3 (Negative pass-through)**.** *We have $\rho_c < 0$ if and only if $\varepsilon_S < 0$ ($\mu_p < 0$) and $|\varepsilon_S| > -\varepsilon_D$.*

Proof. By (18),

$$\rho_c = \frac{\varepsilon_S/\mu_p}{-\varepsilon_D + \varepsilon_S}.$$

Since $\varepsilon_S = S'(\mu)\mu_p\, p/S(\mu)$, the terms ε_S and μ_p have the same sign, so $\varepsilon_S/\mu_p > 0$. Hence $\rho_c < 0$ if and only if

$$-\varepsilon_D + \varepsilon_S < 0,$$

which is equivalent to $\varepsilon_S < 0$ and $|\varepsilon_S| > -\varepsilon_D$. Because ε_S and μ_p have the same sign, $\varepsilon_S < 0$ is equivalent to $\mu_p < 0$. $\square$

The logic is as follows. With $\varepsilon_S < 0$ ($\mu_p < 0$), a higher list price lowers the quantity firms supply, so a leftward shift of supply—caused by a rise in c—creates excess demand at the initial p. Raising the price to eliminate the excess would further contract supply and worsen the imbalance; instead, the price must fall. A decline in p increases both the quantity demanded and, because supply slopes downward, the quantity supplied. When $|\varepsilon_S| > -\varepsilon_D$, the supply curve is steeper than the demand curve, meaning that the supply response to a price cut is larger than the demand response. The price decline therefore eliminates the excess demand and leads to a new equilibrium at a lower list price. This sign reversal cannot occur in the classical model, where $\mu_p = 1$ and supply is everywhere upward-sloping, highlighting a qualitative failure of the no-return benchmark.

At first glance, it may seem puzzling why the price drop—which raises the effective margin ($\mu_p < 0$ and $\rho_c < 0$)—necessarily more than offsets the original one-dollar cost increase. The key is that the price fall must do double duty to clear the market. A one-dollar cost increase shifts the supply curve left, creating an initial excess demand gap. To fill this gap, quantity supplied must increase. Since quantity supplied increases with the effective margin μ, restoring enough supply to cover the original gap already requires μ to rise by one dollar. But the price fall also increases quantity demanded, creating additional excess demand on top of the initial gap. To satisfy both the original supply shortfall and this new demand, supply must increase by more than the original shift, so the effective margin must rise by more than one dollar. Thus, $\mu_p\rho_c > 1$ is precisely the market-clearing requirement when pass-through is negative.

Even when $\mu_p > 0$ and pass-through stays positive, the returns-adjusted formula differs from the no-return case whenever $\mu_p \neq 1$. Comparing the two signs the bias for an analyst who ignores returns. Whether μ_p exceeds one depends on how steeply the return rate falls with price. Differentiating $\mu = p - r(p)(p+h)$ gives

$$\mu_p = 1 - r(p) - r'(p)(p+h).$$

Write $\varepsilon_r(p) \equiv p\, r'(p)/r(p)$ for the elasticity of the return rate in list price. Then $\mu_p > 1$ if and only if $\varepsilon_r(p) < -p/(p+h)$. That is, selection must outweigh the behavioral rise in individual return probabilities by enough that aggregate r falls steeply in p. When instead the behavioral response dominates, ε_r is less negative (or positive) and $\mu_p < 1$.

Result 4. *Relative to the correct returns-adjusted ρ_c, the classical no-return pass-through formula overestimates pass-through when $\varepsilon_r(p) < -p/(p+h)$ and underestimates it otherwise.*

The practical value is that the bias can be signed without estimating the full returns-adjusted model. The return-rate elasticity and the ratio of handling to list price are enough. The economics behind the threshold is familiar from the selection and behavioral forces already in ε_r. When selection dominates, a higher list price lowers the return rate, so the refund-and-handling drag on the effective margin shrinks as price rises. The effective margin can then rise by more than one dollar per dollar of list price, and a smaller price movement closes the excess-demand gap than the classical formula predicts. Costly handling strengthens that channel and makes overestimation more likely. Otherwise, the effective margin rises by less than one dollar per dollar of list price, so list price must move by more than the classical formula predicts to close the gap, and the classical no-return formula underestimates pass-through.

7 Local incidence

Pass-through into list price determines the welfare loss from each cost shifter and how that loss is split between consumers and producers. Throughout, $Q \equiv D(p)$ and $r \equiv r(p)$ denote the equilibrium gross quantity and the return rate. From the consumer-side analysis in Section 3, a small increase in any shifter x changes consumer surplus by

$$\frac{dCS}{dx} = -Q(1-r)\rho_x, \tag{19}$$

where $\rho_x \equiv dp/dx$ is the pass-through of the shock into the list price.

Producer surplus is $PS(\mu, x) = \int_0^\mu S(z,x)\,dz$. Differentiating with respect to x and applying the equilibrium condition $S(\mu, x) = Q$ gives

$$\frac{dPS}{dx} = Q\frac{d\mu}{dx} + \Sigma_x,$$

where $\Sigma_x \equiv \int_0^\mu (\partial S(z,x)/\partial x)\,dz$ captures the direct shift in the area under the supply curve. Expanding the total derivative of the effective margin via the chain rule, $\frac{d\mu}{dx} = \mu_p\rho_x + \mu_x$, yields

$$\frac{dPS}{dx} = Q(\mu_p\rho_x + \mu_x) + \Sigma_x. \tag{20}$$

The local incidence ratio is $I_x \equiv (dCS/dx)/(dPS/dx)$, so

$$I_x = \frac{-Q(1-r)\rho_x}{Q(\mu_p\rho_x + \mu_x) + \Sigma_x}.$$

The components needed to evaluate I_x for each shock are reported in Table 1. Across the four shocks the last column collapses to the same expression in ρ_c and μ_p: the shock-specific pass-through and direct effects cancel in the ratio.

Shock x	ρ_x	μ_x	Σ_x	dCS/dx	dPS/dx	I_x
Production cost c	ρ_c	0	$-Q$	$-Q(1-r)\rho_c$	$Q(\mu_p\rho_c - 1)$	$\frac{(1-r)\rho_c}{1-\mu_p\rho_c}$
Handling cost h	$r\rho_c$	$-r$	0	$-Q(1-r)r\rho_c$	$Qr(\mu_p\rho_c - 1)$	$\frac{(1-r)\rho_c}{1-\mu_p\rho_c}$
Seller tax τ^s	ρ_c	-1	0	$-Q(1-r)\rho_c$	$Q(\mu_p\rho_c - 1)$	$\frac{(1-r)\rho_c}{1-\mu_p\rho_c}$
Buyer tax τ^b	$\mu_p\rho_c$	$-\mu_p$	0	$-Q(1-r)\mu_p\rho_c$	$Q\mu_p(\mu_p\rho_c - 1)$	$\frac{(1-r)\rho_c}{1-\mu_p\rho_c}$

Table 1: Pass-through, direct effects, surplus changes, and incidence ratio for each shock.

Result 5. *For all four shocks, the local incidence ratio is identical:*

$$I_x = \frac{(1-r)\rho_c}{1-\mu_p\rho_c} > 0. \tag{21}$$

Result 5 establishes that the incidence ratio is identical across all four shocks, but the algebra alone can obscure the deeper economic mechanism. The key to understanding this result is that every shock is a scaled perturbation of the same equilibrium geometry. Production cost and seller-tax shocks shift the effective margin μ down by one unit. Handling cost shifts it down by r units. A buyer tax shifts the mapping from list price to μ by μ_p units. Locally, the equilibrium condition $D(p) = S(\mu)$ is linear in the shock's direct effect on μ. The resulting supply shift is proportional to μ_x, and the price adjustment ρ_x that restores market clearing inherits that same proportionality.

Because both $S(\mu)$ and $PS(\mu)$ are locally linear in μ, the entire equilibrium response scales with the shock's effect on the effective margin. The shock's scalar therefore multiplies both consumer and producer surplus changes in the same proportion and cancels out in the incidence ratio. The shock determines how far the equilibrium moves. How that movement is split between consumers and producers depends on the local demand and supply slopes and on local μ_p.[4] Incidence is therefore fixed by market structure, not by which side of the market is hit first. What remains is why that common ratio is positive.

The denominator $1 - \mu_p\rho_c$ in (21) is the per-unit loss in producer surplus. A \$1 cost shock initially reduces producer surplus by Q. The equilibrium list-price rise ρ_c increases the effective margin per unit by $\mu_p\rho_c$. Because $dPS/d\mu = Q$ by the envelope theorem, this marginal gain translates into a total recovery of $Q\mu_p\rho_c$ in producer surplus. The net loss is therefore $Q(1-\mu_p\rho_c)$. The numerator $(1-r)\rho_c$ is the corresponding per-unit consumer loss on kept units. The incidence ratio compares these two per-unit burdens. A small denominator leaves consumers bearing almost the entire shock, while a negative denominator, $\mu_p\rho_c > 1$, means the recovery exceeds the cost, so producers are net gainers. This recovery factor can be written as $\mu_p\rho_c = \varepsilon^S/(\varepsilon^S - \varepsilon^D)$. This expression closely mirrors the pass-through formula, and now we argue that $\rho_c < 0$ if and only if $\mu_p\rho_c > 1$. The condition $\mu_p\rho_c > 1$ requires $\varepsilon^S < 0$, which implies $\mu_p < 0$, and hence $\rho_c < 0$. Conversely, $\rho_c < 0$ requires $\varepsilon^S < 0$ and $|\varepsilon^S| > -\varepsilon^D$, which gives $\mu_p\rho_c > 1$. Therefore, the condition

[4] Consumers care about the list-price change ρ_x. Producers recover only through the effective margin, which a list-price rise moves by $\mu_p\rho_x$. Thus μ_p governs how much of each dollar of p feeds into producer surplus.

governing producer gain from an increase in production cost is exactly the same as that governing negative pass-through.

At first glance, it may seem puzzling why the price drop—which raises the effective margin ($\mu_p < 0$ and $\rho_c < 0$)—necessarily more than offsets the original one-dollar cost increase. The key is that the price fall must do double duty to clear the market. A one-dollar cost increase shifts the supply curve left, creating an initial excess demand gap. To fill this gap, quantity supplied must increase. Since quantity supplied increases with the effective margin μ, restoring enough supply to cover the original gap already requires μ to rise by one dollar. But the price fall also increases quantity demanded, creating additional excess demand on top of the initial gap. To satisfy both the original supply shortfall and this new demand, supply must increase by more than the original shift, so the effective margin must rise by more than one dollar. Thus, $\mu_p\rho_c > 1$ is precisely the market-clearing requirement when pass-through is negative.

The preceding observation implies a striking result: the uniformly positive incidence ratio implies that consumer and producer surplus always move together, regardless of the sign of pass-through. The direction of that common movement depends on both the shock type and the sign of pass-through. When $\rho_c < 0$, production cost, handling cost, and seller tax shocks generate a welfare gain with both surpluses increasing. Under the same condition, buyer-remitted taxes reduce both surpluses. The two tax regimes therefore yield opposite welfare outcomes precisely when pass-through is negative, a sharp violation of classical tax neutrality. When $\rho_c > 0$, all four shocks reduce both surpluses. The central asymmetry is twofold. First, seller and buyer tax remittance can lead to divergent welfare consequences. Second, welfare-improving shocks are possible only for a subset of shocks (cost, handling, seller tax) when pass-through is negative. Neither possibility arises in the no-return benchmark, where pass-through is always positive.

7.1 Deadweight loss and the returns-augmented Harberger triangle

Private surplus is $W = CS + PS$. Summing (19) and (20) gives

$$\frac{dW}{dx} = -Q(1-r)\rho_x + Q(\mu_p\rho_x + \mu_x) + \Sigma_x = Q\big[\mu_x + \big(\mu_p - (1-r)\big)\rho_x\big] + \Sigma_x.$$

Substituting $\mu_p = 1 - r - r'(p)(p+h)$ directly yields

$$\frac{dW}{dx} = Q\big[\mu_x - r'(p)(p+h)\rho_x\big] + \Sigma_x, \tag{22}$$

with μ_x and Σ_x as in Table 1. The term $-Q\,r'(p)(p+h)\rho_x$ is new relative to the classical model. It captures the efficiency consequence of the price-induced change in the return rate. Each returned unit consumes real handling resources h, so a change in the return rate has a first-order welfare impact that is absent when returns are shut down.

Without returns, $r = 0$ and $r' = 0$, so (22) reduces to

$$\frac{dW}{dx} = Q\,\mu_x + \Sigma_x.$$

For a unit production-cost increase, the supply curve shifts vertically upward by one dollar at every quantity. The immediate surplus loss is therefore the area of that shift—a rectangle of height one and width Q. Algebraically, private surplus is $W = CS(p) + PS(p - c)$. Differentiating, the price-change terms $-Q\rho_c$ and $+Q\rho_c$ cancel, leaving $dW/dc = -Q$: a pure resource drain that begins at once.

A unit tax has the same initial impact on private surplus: $W = CS(p) + PS(p-\tau)$ gives $dW/d\tau = -Q$, with the same cancellation of price-change terms. The difference is that the government now collects revenue τQ. Adding it back, total surplus is $V = W + \tau Q$, whose derivative $dV/d\tau = \tau Q'(\tau)$ is zero at $\tau = 0$. Hence an infinitesimal tax creates no net loss; the loss appears only as the tax grows, accumulating quadratically as the classic Harberger triangle $\Delta V \approx -\frac{1}{2}\tau^2 Q'(0)$.

Without returns, cost shocks therefore produce an immediate rectangle of real resource loss, while taxes produce a triangle of distortionary loss that starts from zero. The two are cleanly separated by whether the dollar is lost or merely transferred.

Product returns overturn the clean dichotomy between resource drains and pure transfers that characterizes the textbook model. The return-rate term in (22) is generally nonzero for every shock, so production-cost increases, handling-cost increases, and unit taxes under both remittance regimes all generate first-order welfare changes that cannot be written as mere transfers.

Plugging the values of μ_x and Σ_x from Table 1 into the general welfare expression $dW/dx = Q\big[\mu_x - r'(p)(p+h)\rho_x\big] + \Sigma_x$, and using the pass-through links from Result 2, yields

$$\frac{dW}{dc} = -Q\,\Delta, \qquad \frac{dW}{dh} = -Q\,r\,\Delta, \qquad \frac{dW}{d\tau^s} = -Q\,\Delta, \qquad \frac{dW}{d\tau^b} = -Q\,\mu_p\,\Delta, \tag{23}$$

where the common factor is defined as

$$\Delta \equiv 1 + r'(p)(p+h)\,\rho_c. \tag{24}$$

The unit term in Δ is the mechanical welfare effect that would operate even if returns did not respond to price, while $r'(p)(p+h)\rho_c$ captures the additional efficiency consequence of the price-induced reallocation between kept and returned units—each percentage-point change in the return rate, $r'(p)\rho_c$, imposes a real resource cost of $p+h$ per unit sold.

The sign of Δ governs the direction of welfare effects for all shocks except the buyer-tax scaling. Since the incidence analysis established that consumer and producer surplus always move together, $\Delta < 0$ exactly when $\rho_c < 0$. The welfare implications that follow are exactly those already discussed in the incidence section. The magnitude of these effects, not just their sign, also carries economic content. From (24), that magnitude is large when the behavioral term $r'(p)(p+h)\rho_c$ is large in absolute value.

When $\rho_c > 0$, two distinct regimes are possible. If $\mu_p > 0$, the supply curve slopes upward in list-price space, so a cost increase raises p. Whenever $r'(p) > 0$, that price increase raises the return rate, adds handling costs, and pushes Δ above one. Whenever $r'(p) < 0$, it lowers the return rate, partially offsets the mechanical loss, and yields $\Delta < 1$. The welfare loss $Q\Delta$ is therefore larger when $r'(p)$ is positive and large, because each dollar of pass-through raises returns and wastes an additional $p+h$ per unit sold.

If instead $\mu_p < 0$ (which requires $|\varepsilon_S| < -\varepsilon_D$), supply itself slopes downward in list-price space. A higher p lowers μ and therefore contracts quantity supplied as well as quantity demanded. Pass-through remains positive because demand is more price-responsive than this downward-sloping supply, so a price increase still reduces excess demand. The return-rate response must already be steep in this region, since $\mu_p < 0$ requires $r'(p)(p+h) > 1 - r$. Even a modest ρ_c can then make the behavioral contribution $r'(p)(p+h)\rho_c$ large relative to the mechanical unit term, and thus leave Δ substantially above one. A small price rise still reallocates many purchases into costly returns.

When $\rho_c < 0$, necessarily $\Delta < 0$, so cost or seller-tax increases raise private surplus by $Q|\Delta|$. The gain is larger the more negative is Δ, which occurs when $r'(p)(p+h)\,|\rho_c|$ is large. Because $\rho_c < 0$, the cost increase lowers the list price, which reduces the return rate and saves $p+h$ on each unit that switches from return to keep. Those savings can exceed the direct resource cost of the shock, generating a Pareto improvement. The gain is amplified when μ_p is very negative and the behavioral response $r'(p)$ is strong, so that a small downward price adjustment delivers large handling-cost savings.

A similar discussion translates into a comparison with the no-return benchmark, which predicts a first-order private-surplus loss of exactly Q for a unit production-cost or seller-tax increase. The correct returns-adjusted change is $dW/dc = -Q\Delta$, so the classical rectangle over- or understates the true effect according to whether Δ lies below or above one.

Result 6 (Bias from ignoring returns in first-order welfare)**.** *Consider a unit production-cost or seller-tax increase. Relative to the correct returns-adjusted change $dW/dc = -Q\Delta$, the no-return formula $dW/dc = -Q$*

(i) underestimates the welfare loss if and only if $\rho_c > 0$ and $r'(p) > 0$ (hence $\Delta > 1$);

(ii) overestimates the welfare loss if and only if $r'(p) < 0$ (which implies $\mu_p > 0$ and hence $\rho_c > 0$, so $0 < \Delta < 1$);

(iii) predicts a loss of Q when private surplus in fact rises if and only if $\rho_c < 0$, in which case the rise is $Q|\Delta|$.

For a buyer-remitted tax the first-order change is instead $dW/d\tau^b = -Q\mu_p\Delta$. Compared with the no-return benchmark $-Q$, this scaling by $\mu_p\Delta$ gives the following corollary.

Corollary 1 (Bias under buyer remittance)**.** *Relative to the no-return prediction $dW/d\tau = -Q$, a unit buyer-remitted tax satisfies $dW/d\tau^b = -Q\mu_p\Delta$.*

(i) If $\mu_p > 0$ (which implies $\rho_c > 0$ and hence $\Delta > 0$), the no-return formula underestimates the welfare loss if and only if $\mu_p\Delta > 1$ and overestimates it if and only if $0 < \mu_p\Delta < 1$.

(ii) If $\mu_p < 0$, the no-return formula predicts a loss of Q, whereas private surplus rises by $Q\,|\mu_p|\Delta$ if and only if $\rho_c > 0$ and falls by $Q\,|\mu_p\Delta|$ if and only if $\rho_c < 0$.

Appendix A restates these cases in return and supply primitives—the form most useful for empirical checks.

A Welfare bias in primitives

Recall $\varepsilon_r(p) \equiv p\,r'(p)/r(p)$ and $\mu_p = 1-r(p)-r'(p)(p+h)$. Then $r'(p)(p+h) = \varepsilon_r(p)\,r(p)\,(p+h)/p$, so

$$\mu_p > 0 \iff \varepsilon_r(p) < \frac{p}{p+h}\,\frac{1-r(p)}{r(p)}, \tag{25}$$

$\mu_p < 0$ is the reverse strict inequality, and ε_r has the same sign as r' (throughout, $r(p) \in (0,1)$).

Result 6 becomes:

(i) underestimation iff $\varepsilon_r(p) > 0$ and either $\mu_p > 0$, or both $\mu_p < 0$ and $|\varepsilon_S| \le -\varepsilon_D$;

(ii) overestimation iff $\varepsilon_r(p) < 0$;

(iii) wrong sign (predicted loss, actual gain) iff $\mu_p < 0$ and $|\varepsilon_S| > -\varepsilon_D$.

Proof. By Result 6, underestimation is $\rho_c > 0$ with $r' > 0$, overestimation is $r' < 0$, and the wrong-sign case is $\rho_c < 0$. Since ε_r has the same sign as r', write these in ε_r. Result 3 translates the sign of ρ_c into conditions on μ_p and $|\varepsilon_S| \gtrless -\varepsilon_D$: with $\varepsilon_r > 0$ this is (i), and $\rho_c < 0$ is (iii). For (ii), $\varepsilon_r < 0$ forces $\mu_p > 1-r > 0$ and hence $\rho_c > 0$, so $0 < \Delta < 1$. □

Corollary 1 becomes:

(i) If $\mu_p > 0$, underestimation iff $\dfrac{\varepsilon_S - \varepsilon_D}{\varepsilon_D} > \varepsilon_r(p)\dfrac{p+h}{p}$, and overestimation iff $\dfrac{\varepsilon_S - \varepsilon_D}{\varepsilon_D} < \varepsilon_r(p)\dfrac{p+h}{p}$;

(ii) If $\mu_p < 0$, surplus rises iff $|\varepsilon_S| \le -\varepsilon_D$ and falls iff $|\varepsilon_S| > -\varepsilon_D$.

Proof. Under $\mu_p > 0$ we have $\varepsilon_S > 0$, so $\varepsilon_S - \varepsilon_D > 0$, and also $\rho_c > 0$, $\Delta > 0$. Write $\mu_p = 1 - r(p) - r'(p)(p+h)$ and $\Delta = 1 + r'(p)(p+h)\rho_c$. Using $\mu_p\rho_c = \varepsilon_S/(\varepsilon_S - \varepsilon_D)$,

$$\begin{aligned}
\mu_p\Delta - 1 &= \mu_p\big(1 + r'(p)(p+h)\rho_c\big) - 1 \\
&= \mu_p - 1 + r'(p)(p+h)\,(\mu_p\rho_c) \\
&= -r(p) - r'(p)(p+h) + r'(p)(p+h)\,\frac{\varepsilon_S}{\varepsilon_S - \varepsilon_D} \\
&= -r(p) + r'(p)(p+h)\,\frac{\varepsilon_D}{\varepsilon_S - \varepsilon_D} \\
&= -r(p)\left(1 - \varepsilon_r(p)\,\frac{p+h}{p}\,\frac{\varepsilon_D}{\varepsilon_S - \varepsilon_D}\right).
\end{aligned}$$

Hence $\mu_p\Delta > 1$ iff $\varepsilon_r(p)\frac{p+h}{p}\frac{\varepsilon_D}{\varepsilon_S - \varepsilon_D} > 1$. Since $\varepsilon_D/(\varepsilon_S - \varepsilon_D) < 0$, multiplying through reverses the inequality and yields the underestimation cut in (i). The overestimation cut is the reverse; under $\mu_p > 0$ and $\Delta > 0$ it is exactly $0 < \mu_p\Delta < 1$. Part (ii): under $\mu_p < 0$, surplus rises iff $\rho_c > 0$ and falls iff $\rho_c < 0$, which is Result 3. Regions may be read in ε_r via (25). □